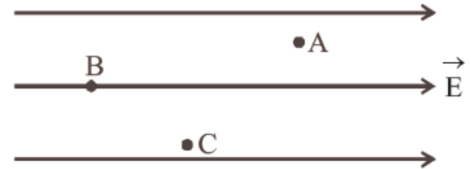


Section A - MCQ

Q1. On decreasing the distance between the plates of a parallel plate capacitor, its capacitance [1]

- A. remains unaffected C. first increases then decreases
B. decreases D. increases

Q2. A, B and C are three points in a uniform electric field. The electric potential is [1]



- A. maximum at B C. same at all the three points A, B and C
B. maximum at C D. maximum at A

Q3. A and B are two points in an electric field. If the work done in carrying 4 C of electric charge from A to B is 16 J, the potential difference between A and B is [1]

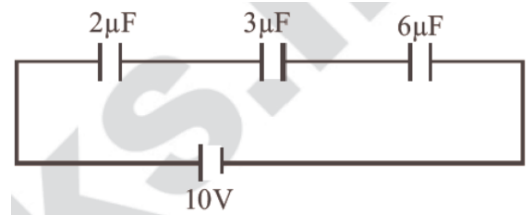
- A. Zero B. 2 V C. 4 V D. 16 V

Q4. To obtain $3 \mu F$ capacity from three capacitors of $2 \mu F$ each, they will be arranged [1]

- A. all the three in series
B. all the three in parallel
C. two capacitors in series and the third in parallel with the combination of the first two
D. two capacitors in parallel and the third in series with the combination of first two

Q5. In the given figure, the charge on $3 \mu F$ capacitor is [1]

- A. $10 \mu F$ C. $30 \mu F$
B. $15 \mu F$ D. $5 \mu F$



Q6. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential at a distance of 2 cm from the centre of the sphere is [1]

- A. Zero B. 10 V C. 4 V D. $10/3$ V

Q7. The electric potential on the axis of an electric dipole at a distance ' r ' from it's centre is V. Then the potential at a point at the same distance on its equatorial line will be [1]

- A. $2V$ B. $-V$ C. $\frac{V}{2}$ D. Zero

Q 8. At a point A, there is an electric field of 500 V/m and potential difference of 3000 V. The distance between the point charge and A is. [1]

- A. 6 m B. 36 m C. 12 m D. 144 m

Q 9. An electric dipole having dipole moment of 4×10^{-9} C m is placed in a uniform electric field such that the dipole is in stable equilibrium. If the magnitude of the electric field is 3×10^3 N/C, What is the work done in rotating the dipole to a position of unstable equilibrium? [1]

- A. Zero B. 1.2×10^{-5} J C. 2.4×10^{-5} J D. -1.2×10^{-5} J

Q 10. The distance between the two plates of a parallel plate capacitor is doubled and the area of each plate is halved. If C is its initial capacitance, its final capacitance is equal to [1]

- A. 2C B. $\frac{C}{2}$ C. 4C D. $\frac{C}{4}$

Q 11. Why are electric field lines are perpendicular at a point on an equipotential surface of a conductor? [1]

Q 12. What is an equipotential surface. Sketch equipotential surfaces for a dipole. [1]

Section B - Very Short Answer Questions

Q 13. What is the area of the plates of a 2 Farad parallel plate air capacitor, given that the separation between the plates is 0.5 cm? [2]

Q 14. An air capacitor is given a charge of $2\mu\text{C}$ raising its potential to 200 V. If on inserting a dielectric medium, its potential falls to 50 V. What is the dielectric constant of the medium. [2]

Q 15. (a) Calculate the potential at a point P due to a charge of $4 \times 10^{-7}\text{C}$ located 9 cm away. [2]
(b) Hence obtain the work done in bringing a charge of 2×10^{-9} C from infinity to the point P. Does the answer depend on the path along which the charge is brought?

Q 16. A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. [2]
(a) Calculate the capacitance of this capacitor.
(b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? (The value of $\epsilon_0 = 8.85 \times 10^{-12} \text{ Nm}^2\text{C}^{-2}$)

Section C - Short Answer Questions

Q 17. Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in vacuum in terms of plate area A and separation d between the plates. [3]

Q 18. A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A. [3]

Q 19. Two charges 5×10^{-8} C and -3×10^{-8} C are located 16 cm apart. At what point(s) on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero [3]

- Q 20. (a) Define potential energy of a system of two charges. [3]
 (b) Two-point charges q_1 and q_2 , separated by a distance r_{12} are kept in an external electric field. Derive an expression for the potential energy of the system of two charges in the field.

Section D - Long Answer Questions

- Q 21. A parallel plate capacitor of capacitance C is charged to a potential V by a battery. Without disconnecting the battery, the distance between the plates is tripled and a dielectric medium of $k = 10$ is introduced between the plates of the capacitor. Explain giving reasons, how will the following be affected: [4]
 (i) capacitance of the capacitor (ii) charge on the capacitor.
- Q 22. A parallel plate capacitor, of capacitance 20 pF , is connected to a 100 V supply. After sometime the battery is disconnected, and the space, between the plates of the capacitor is filled with a dielectric, of dielectric constant 5 . Calculate the energy stored in the capacitor [4]
 (i) before
 (ii) after the dielectric has been put in between its plates.
- Q 23. (a) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (i) maximum, and (ii) minimum? [5]
 (b) An electric dipole consist of point charges -1 pC and $+1 \text{ pC}$ located at $(0, 0)$ and $(3 \text{ mm}, 4 \text{ mm})$ respectively in x-y plane. An electric field $\vec{E} = \left(\frac{1000 \text{ V}}{m}\right) \hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.
- Q 24. (a) An electric dipole (dipole moment $\vec{p} = p\hat{i}$), consisting of charges $-q$ and q , separated by distance $2a$, is placed along the x-axis, with its centre at the origin. Show that the potential V , due to this dipole, at a point x , ($x \gg a$) is equal to $\frac{1}{4\pi\epsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$. [5]
 (b) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) \text{ Cm}^{-2}$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1 .